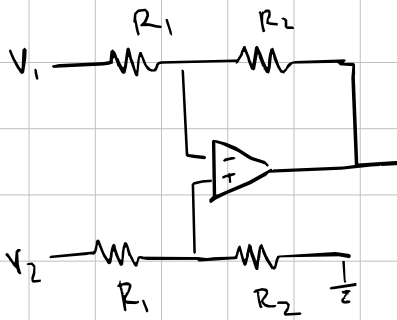
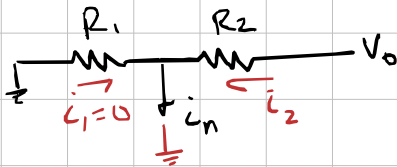


①



This page for -4050 only

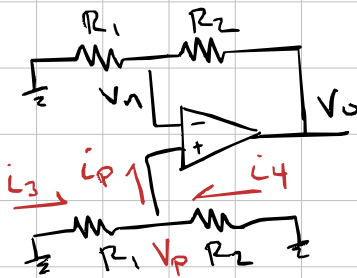
$i_n: V_p = 0, V_n = 0$



$$i_n = \cancel{i_1} + i_2 = i_2$$

$$V_0^A = i_n R_2$$

$i_p:$



$$i_p = i_3 + i_4 = (0 - V_p)/R_1 + (0 - V_p)/R_2$$

$$i_p = -V_p \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = -V_p \left(\frac{1}{R_1 \parallel R_2} \right)$$

$$i_p = -V_p \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$

$$V_p = -i_p \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$

$$V_0 = - \left(1 + \frac{R_2}{R_1} \right) \left(\frac{R_1 R_2}{R_1 + R_2} \right) i_p$$

$$V_0 = - \left(\frac{R_1 + R_2}{R_1} \right) \left(\frac{R_1 R_2}{R_1 + R_2} \right) i_p$$

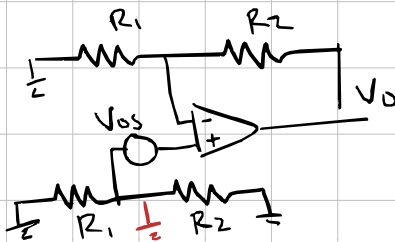
$$V_0^B = -i_p R_2$$

$$V_0 = V_0^A + V_0^B = i_n R_2 - i_p R_2 = R_2 (i_n - i_p)$$

$$V_0 = R_2 (I_B - I_{os}/2 - I_B - I_{os}/2) = \underline{\underline{I_{os} R_2 = V_0}}$$

②

V_{os} from V_0 error:



→ effectively non-inv. Amp
for $V_{os} \rightarrow A = 1 + \frac{R_2}{R_1}$

$$\text{Effective } V_{os} \rightarrow V_0 = A V_{os} \rightarrow I_{os} R_2 = \left(1 + \frac{R_2}{R_1} \right) V_{os}$$

$$V_{os} = \frac{R_1 R_2}{R_1 + R_2} I_{os} = R_1 \parallel R_2 I_{os}$$

③

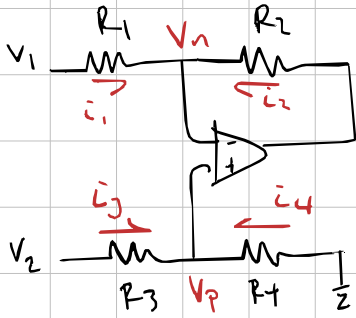
$$I_B = 450 \text{ nA}, I_{os} = 50 \text{ nA} \rightarrow \pm 50 \text{ nA}$$

$$V_0 = \pm 50 \text{ nA} (100 \text{ k}\Omega)$$

$$\underline{\underline{-5 \text{ mV} \leq V_0 \leq 5 \text{ mV}}}$$

①

This page for -6050 only



$$\rightarrow \bar{i}_n: V_n = 0 \rightarrow V_o = \bar{i}_n R_2$$

$$\bar{i}_p: V_i = 0$$

$$i_p = \bar{i}_3 + i_4 = -V_p/R_3 - V_p/R_4$$

$$i_p = -V_p \left(\frac{1}{R_3} + \frac{1}{R_4} \right)$$

$$V_p = -i_p R_3 \parallel R_4$$

Gain of $V_p \rightarrow$ Non-inv. Amp.

$$V_o = V_p \left(1 + \frac{R_2}{R_1} \right) = -i_p \frac{R_3 R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4}$$

$$V_o = \bar{i}_n R_2 - i_p \frac{R_3 R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4}$$

$$V_o = R_2 (I_B - I_{os}/2) - \frac{R_3 R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4} (I_B + I_{os}/2)$$

$$V_o = \left[R_2 - \frac{R_3 R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4} \right] I_B - \left[R_2 + \frac{R_3 R_4}{R_1} \frac{R_1 + R_2}{R_3 + R_4} \right] I_{os}/2$$

②

if $R_4 = 2R_2$: (I_B term only)

$$R_2 - \frac{2R_3 R_2}{R_1} \frac{R_1 + R_2}{R_1 + 2R_2} = 0$$

$$R_2 \left(1 - 2 \frac{R_3}{R_1} \frac{R_1 + R_2}{R_1 + 2R_2} \right) = 0$$

$\rightarrow$ if $R_2 = 0$, I_B term $\rightarrow 0$
or...

$$\frac{R_3 + R_2}{2} - R_3 - \frac{R_3 R_2}{R_1} = 0$$

$$R_2 - R_3/2 - R_3 R_2/R_1 = R_2 - R_3 \left(\frac{1}{2} + \frac{R_2}{R_1} \right) = 0$$

$$R_3 = R_2 / \left(\frac{1}{2} + \frac{R_2}{R_1} \right) = \frac{1}{\frac{1}{2R_2} + \frac{1}{R_1}} \rightarrow R_3 = R_1 \parallel 2R_2$$

either acceptable

③

$$I_B = 450 \text{ nA}, I_{os} = 50 \text{ nA} \rightarrow \pm 50 \text{ nA}$$

$$R_3 = R_1, R_4 = R_2$$

$$V_o = \left[R_2 - \frac{R_1 R_2}{R_1} \frac{R_1 + R_2}{R_1 + R_2} \right] I_B - \left[R_2 + \frac{R_1 R_2}{R_1} \frac{R_1 + R_2}{R_1 + R_2} \right] I_{os}/2$$

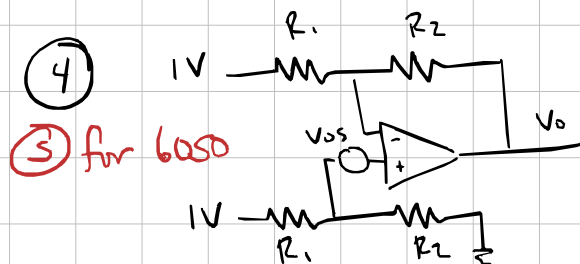
$$= R_2 - R_2 = 0$$

$$= R_2 + R_2 = 2R_2$$

$$V_o = -R_2 I_{os}$$

$$V_o = \pm 50 \text{ nA} (100 \text{ k}\Omega)$$

$$-5 \text{ mV} \leq V_o \leq 5 \text{ mV}$$



4 error components: ① Voltage offset @ 25°C

② temperature change

③ CMRR ④ PSRR

2. Temperature change: Max swing from 25°C is to $0^\circ\text{C} \rightarrow \Delta T = 25^\circ\text{C}$ (or -25°C)

3. V_{cm} ? calculate V^+ & assume virtual short:

$$V_{cm} = V^+ = \frac{R_2}{R_1 + R_2} 1V = 0.9901V$$

$$CMRR_{dB} = 80dB \rightarrow CMRR = 10000\%$$

4. Power range: both rails have $\pm 0.2V$.

therefore, $\Delta V_{supply} = 0.2V$ (or $0.4V$, depending on original voltage reference)

$$PSRR_{dB} = 86dB \rightarrow PSRR = 19953\%$$

$$\max V_{os} = 1mV + (5\mu V/^\circ\text{C})(25^\circ\text{C}) + \frac{0.9901V}{10000\%} + \frac{0.2V}{19953\%}$$

$$\max V_{os} = 1mV + 125\mu V + 99\mu V + 10\mu V$$

$$\max V_{os} = 1.234mV$$

$$\max V_o \text{ error} = \max V_{os} \cdot \left(1 + \frac{R_2}{R_1}\right)$$

$$\max V_o \text{ error} = 124.6mV$$

⑤

$$A(jf) = a_0 / (1 + jf/f_b)$$

known: $a_0 = 80 \text{ dB} = 10^4 \text{ V/V}$

$$\text{GBP} = 50 \text{ MHz} = f_t$$

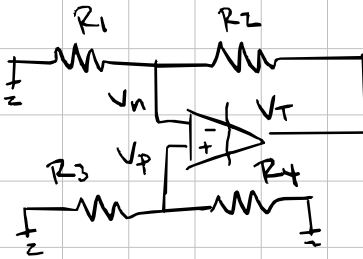
$$f_t = a_0 f_b \rightarrow f_b = f_t / a_0$$

$$f_b = 5 \text{ kHz}$$

$$A(jf) = \frac{10000 \text{ V/V}}{1 + jf/5000 \text{ Hz}}$$

⑥

$\beta \rightarrow$ set sources to 0, $\beta = -\frac{V_D}{V_T}$



$$V_p = 0 \quad V_n = \frac{R_1}{R_1 + R_2} V_T$$

$$V_D = -V_n$$

$$\rightarrow \beta = \frac{R_1}{R_1 + R_2}$$

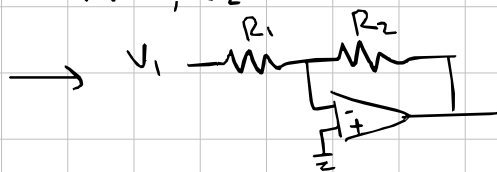
$$f_B = f_b (1 + a_0 \beta) = 5000 (1 + 10000 \cdot \frac{1}{101})$$

$$f_B = 500 \text{ kHz}$$

⑦ for 6050

⑦ Asking for superposition development of gain...

⑧ for 6050
 $A_1(jf) \rightarrow V_1 \neq 0, V_2 = 0$

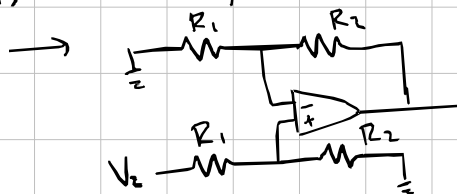


$$A_{ideal} = -\frac{R_2}{R_1}$$

Assume $A_0 = A_{ideal}$
 (no low-f gain error)

$$A_1(jf) = -\frac{R_2/R_1}{1+jf/f_B} = -100 / (1+jf/500\text{kHz})$$

$A_2(jf) \rightarrow V_1 = 0, V_2 \neq 0$



$$V_P = \frac{R_2}{R_1 + R_2} V_2$$

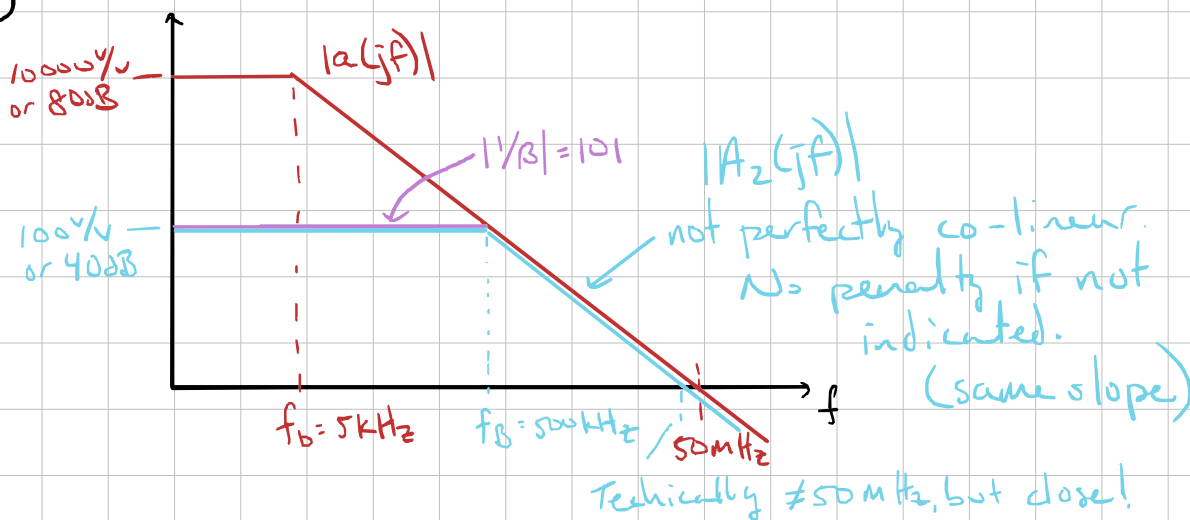
$$V_O = \left(1 + \frac{R_2}{R_1}\right) \left(\frac{R_2}{R_1 + R_2}\right) V_2$$

$$V_O = \frac{R_2}{R_1} V_2 \rightarrow A_{ideal} = \frac{R_2}{R_1}$$

$$A_2(jf) = \frac{R_2/R_1}{1+jf/f_B} = 100 / (1+jf/500\text{kHz})$$

⑨ for 6050

⑧

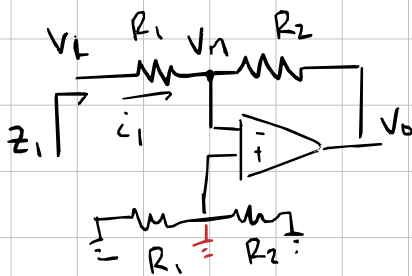


⑨ Z_1 : (same as inverting amp)

⑩ for 6050

Again: $V_2 = 0$, $V_1 \neq 0 \rightarrow A_1(jf)$

(Done Directly,
alternatively done with
Blackman's...
next page)



$$Z_1 = V_1 / i_1$$

$$V_1 - V_n = i_1 R_1$$

$$\text{with } A_1 = -\frac{R_2}{R_1} / (1 + jf/f_B)$$

then V_n is...

$$V_n = (V_0 - V_1) \frac{R_1}{R_1 + R_2} + V_1 = (A_1 V_1 - V_1) \frac{R_1}{R_1 + R_2} + V_1$$

$$\rightarrow V_1 - V_n = i_1 R_1 = V_1 (1 - A_1) \frac{R_1}{R_1 + R_2}$$

$$i_1 = \frac{V_1}{R_1 + R_2} (1 - A_1)$$

$$Z_1 = \frac{R_1 + R_2}{(1 - A_1)}$$

$$1 - A_1 = \frac{1 + jf/f_B}{1 + jf/f_B} - \frac{-R_2/R_1}{1 + jf/f_B} = \frac{1 + \frac{R_2}{R_1} + jf/f_B}{1 + jf/f_B} = \left(1 + \frac{R_2}{R_1}\right) \frac{1 + jf/f_1}{1 + jf/f_B}$$

$$Z_1 = \frac{R_1 + R_2}{1 + \frac{R_2}{R_1}} \frac{1 + jf/f_B}{1 + jf/f_1} = R_1 \frac{1 + jf/f_B}{1 + jf/f_1}$$

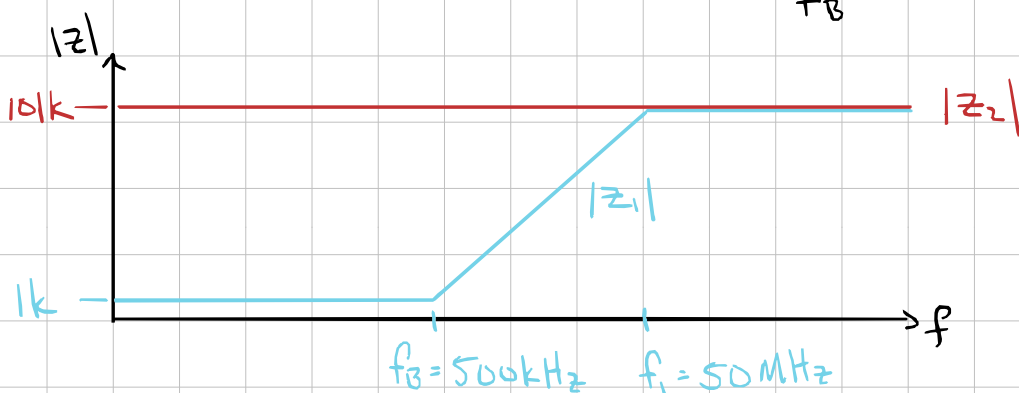
$$f_1 = f_B \left(1 + \frac{R_2}{R_1}\right) = 50.5 \text{ MHz} \approx f_t$$

Z_2 : Assuming $r_d \rightarrow \infty$, then impedance is just:

$$Z_2 = R_1 + R_2$$

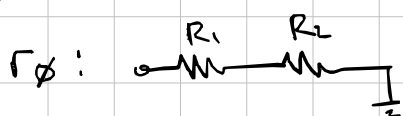
limits: $Z_1(f \rightarrow 0) = R_1 = 1 \text{ k}\Omega$

$$Z_1(f \rightarrow \infty) = R_1 \frac{f_B (1 + \frac{R_2}{R_1})}{f_B} = R_1 + R_2 = 101 \text{ k}\Omega$$

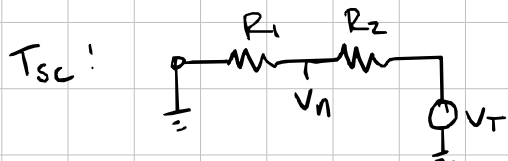


⑨ using Blackman's formula for Z_1

⑩ for 6050

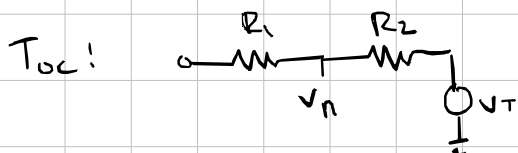


$$r_\phi = R_1 + R_2$$



$$V_n = V_T \frac{R_1}{R_1 + R_2}$$

$$T_{sc} = a(jf) \beta_{sc} = a(jf) \frac{R_1}{R_1 + R_2}$$



No current through either R...

$$V_n = V_T$$

$$T_{oc} = a(jf)$$

$$Z_1 = R_1 + R_2 \left[1 + a_0 \frac{R_1}{R_1 + R_2} / 1 + jf/f_b \right] / \left[1 + a_0 / 1 + jf/f_b \right]$$

$$Z_1 = R_1 + R_2 \left[1 + jf/f_b + a_0 \frac{R_1}{R_1 + R_2} \right] / \left[1 + jf/f_b + a_0 \right]$$

$$Z_1 = \underline{(R_1 + R_2) \left(1 + a_0 \frac{R_1}{R_1 + R_2} \right) \left(\frac{1}{1 + a_0} \right)} \left[1 + jf/f_z \right] / \left[1 + jf/f_p \right]$$

$$= (R_1 + R_2 + a_0 R_1) / (1 + a_0)$$

$$f_z = f_b (1 + a_0 \frac{R_1}{R_1 + R_2}) = f_B$$

$$= (R_1(1 + a_0) + R_2) / (1 + a_0)$$

$$f_p = f_b (1 + a_0) \approx f_t$$

$$= R_1 + R_2 / (1 + a_0)$$

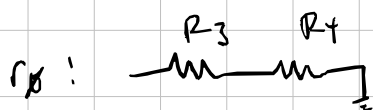
$$f \rightarrow 0 \quad Z_1 = R_1 + R_2 / 1 + a_0 \approx R_1$$

$$f \rightarrow \infty \quad Z_1 = (R_1 + R_2 / 1 + a_0) \left[f_b (1 + a_0) / f_b (1 + a_0 \frac{R_1}{R_1 + R_2}) \right]$$

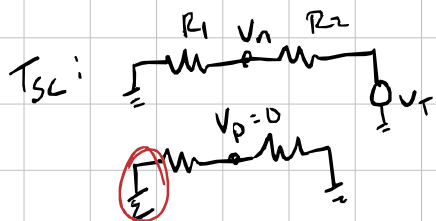
$$= (R_1 + R_2 / 1 + a_0) (1 + a_0) / (1 + a_0 \frac{R_1}{R_1 + R_2})$$

$$\approx R_1 \quad a_0 / a_0 \frac{R_1}{R_1 + R_2} \approx R_1 + R_2$$

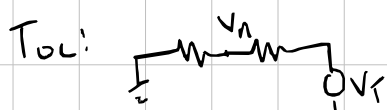
Repeat for Z_2 for completeness:



$$r_\phi = R_3 + R_4 = R_1 + R_2$$



$$V_n = V_T \frac{R_1}{R_1 + R_2} \quad T_{sc} = a(jf) \frac{R_1}{R_1 + R_2}$$



$$\text{Same as } T_{sc} \Rightarrow T_{oc} = T_{sc}$$

$$Z_2 = R_1 + R_2 \frac{1 + T_{sc}}{1 + T_{sc}} = R_1 + R_2$$

⑩ full power Bandwidth: ← assume 5V from Q4?

⑪ for 6050

$$f_{PB} \leq SR / 2\pi V_{sat}$$

$$f_{PB} \leq 35 \text{ V}/\mu\text{s} / 2\pi 5\text{V} = 1.11 \text{ MHz}$$

since $f_B < f_{PB}$, then small-signal limited.

⑪ Op-Amp is CFA $\rightarrow$ Get closed-loop Gain, show still Diff. Amp. $\rightarrow V_o = z i_n$

⑫ for 6050



$$i_n = i_1 + i_2 = (V_p - V_1) / R_1 + (V_p - V_o) / R_2 = V_o / z$$

$$\frac{V_z}{R_1 + R_2} \left(\frac{R_2}{R_1} + 1 \right) - \frac{V_1}{R_1} = \frac{V_o}{z} + \frac{V_o}{R_2}$$

$$\frac{R_2}{R_1} (V_z - V_1) = V_o \left(\frac{R_2}{z} + 1 \right)$$

$$V_o = \frac{1}{1 + R_2/z} \frac{R_2}{R_1} (V_z - V_1)$$

↑ frequency Dependence gain Difference